Cold Weather Protection – General Data

1. General

CAUTION **SAE AEROSPACE MATERIALS SPECIFICATION (AMS) 1428; "AIRCRAFT DE-ICING/ANTI-ICING FLUID SAE TYPE II AND IV" ARE NOT ACCEPTABLE.**

Operating the aircraft in cold weather conditions presents several problems due to the effects of low temperature, such as ice and snow buildup on wings and control surfaces. Corrective maintenance and lost time can be kept to a minimum by an organized program of preventative maintenance. Ensure aircraft parking is observed in accordance with procedures detailed in 10-00-00, Para 1.

Note Procedures given may be varied according to conditions and available equipment. In extremely cold weather, covers can be prevented from freezing to the aircraft by applying a light brush coat of freezing point depressant fluid.

Equipment and Materials:

- Long handled broom
- Rig, De-icer/Washer FMC 239–5500, Model SA–600, or equivalent
- SAE Aerospace Materials Specification (AMS) 1424; "Aircraft De-icing/Anti-icing Fluid SAE Type I"
- SAE Aerospace Materials Specification (AMS) 1428; "Aircraft De-icing/Anti-icing Fluid SAE Type III"

2. Aircraft De-icing

Aircraft de-icing is defined as the process of removing accumulation of snow or ice from the aircraft. Where facilities exist for heating fluids contained in the de-icer rig, note that heated fluids are more effective in the de-icing procedure than unheated fluids. Remove snow and ice from aircraft as follows:

A. Remove snow from fuselage, wings and empennage as follows:

1) Immediately after a snow fall, clean the aircraft of all snow to prevent any melting snow from freezing to the surface and interior areas of the aircraft into which water may flow.

CAUTION **ALL EXTERIOR SURFACES, PARTICULARLY AIRFOIL SURFACES, MUST BE COMPLETELY CLEAR OF SNOW, SLUSH OR ICE IMMEDIATELY PRIOR TO TAKE-OFF.**

2) Wet snow can best be removed from the fuselage, wings and empennage by means of a long handled broom if depth of snow warrants it, and by application of freezing point depressant (FPD) fluids. When sweeping snow off aircraft surfaces, work from support stands placed close to the aircraft. If stands are not available, exercise extreme caution when walking on aircraft surfaces during snow clearing operations to avoid slipping or sliding off aircraft. Wear rubber or fabric footwear while on aircraft surfaces.

Note FPD fluids must be applied at the correct mixture strength and temperature as determined by manufacturer.

3) Check for snow accumulation between aileron leading edges and flaps, elevator leading edges and shrouds, wing flap leading and trailing edges and all tabs and shrouds.

4) If snow accumulation is dry and light, it may be blown off the aircraft using compressed air.

Note Do not use hot air to remove dry snow.

5) Clear aircraft parking area of snow, slush and ice. If this is impractical, clear the area around the main wheels and nose wheels to reduce the possibility of tires being frozen to the ground.

B. Remove ice accumulation from aircraft as follows:

1) Plug engine and air intake ducts when de-icing is done in their vicinity.

CAUTION **USE FREEZING POINT DEPRESSANT (FPD) FLUIDS WITH CAUTION NEAR ENGINES AND EXHAUST GAS AREAS TO AVOID ACCUMULATIONS THAT COULD BE INGESTED OR IGNITED ON STARTING ENGINES. REMOVE ANY ACCUMULATION BEFORE STARTING ENGINES.**

2) Drain holes, static ports, pitot heads, area between inboard fore flap trailing edge and trailing flap leading edge, outboard flap trailing edge and aileron leading edge, and control surfaces must be free of ice accumulation.

CAUTION **DO NOT USE HARD OR SHARP TOOLS FOR REMOVING ICE. USE FPD FLUIDS SPARINGLY AROUND HINGE POINTS AND BEARINGS. DO NOT SPRAY DIRECTLY ONTO THESE AREAS.**

3) Remove ice from the aircraft surface by spray application of FPD fluids; as the ice loosens, sweep it off.

Note Application of hot air can also be used to melt ice from surfaces but do not direct hot air on window surfaces. Melting ice must be mopped dry to prevent subsequent refreezing on aircraft.

4) When operating from snow or slush covered runways, monitor the nose gear area externally for ice buildup, and if necessary remove any ice buildup by spray application of FPD fluids.

3. Aircraft Anti-icing

A. General

Anti-icing is defined as the procedure used to prevent or delay the formation of ice accumulations on the aircraft and flight control surfaces. Application of unheated freezing point depressant (FPD) fluids is more effective in the anti-icing procedures than heated fluids. Carry out anti-icing as follows:

1) Apply FPD fluids at the correct mixture as determined by the manufacturer.

2) Ensure spray application of FPD fluid covers all surfaces with a fine film of fluid.

Note Complete access must be gained to upper surfaces of horizontal stabilizer and elevators.

B. Elevator Assembly Anti-icing

1) Prevent ice accumulation at the elevator trim tabs by spraying FPD fluid on to the tab leading edge and around hinge locations.

2) Additional anti-icing protection of the elevator can be achieved by the periodic application of silicone grease to the tab leading edges.

Note Two operators are required to carry out this task. Repeat silicone grease application as frequently as weather conditions demand.

If the aircraft is washed (particularly the horizontal stabilizer area) or de-iced while operating in climatic conditions where freezing or ice buildup could occur, repeat the silicone grease application to the tabs.

3) De-ice horizontal stabilizer and elevator surfaces if required.

4) Engage elevator gust lock.

5) Ensure all tabs are free of snow and ice accumulation.

6) Operate trim tabs to fully up position. Using a clean cotton cloth dampened with isopropyl alcohol, clean the full length of the leading edge tabs; clean as much of the leading edge area as possible, dry immediately with a clean dry cloth.

7) Operate trim tabs to fully down position; repeat cleaning as at Para 3.B.6).

CAUTION EXERCISE CARE WHEN APPLYING SILICONE GREASE.

8) Using a soft bristle brush, apply a thin film of silicone grease to the full length of the leading edge of the tabs; cover as much of the leading edge as possible.

9) Operate trim tabs to full up positions; repeat application of silicone grease as in Para 3.B.8).

10) Disengage elevator gust lock.

11) Operate trim tabs to take-off position.

C. Propeller Anti-icing

1) To help prevent ice accumulation on propeller blades, Icex is applied to blades.

2) Lightly scrub blades and heaters with a mild soap and water solution and dry thoroughly.

3) Apply Icex with a soft cloth or aerosol can sparingly over blades.

4) Buff the entire surface with a soft dry cloth until a smooth gloss is obtained.

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